RESTAURANT CLOSURE ANALYSIS

What Predicts Whether a Restaurant Survives?

A SQL & Power BI case study on 49,853 U.S. restaurants

Detail	Summary
Data source	Yelp Open Dataset (business.json)
Final scope	49,853 restaurants · 13 U.S. states · 903 cities
Tools	Python (extraction & cleaning) · PostgreSQL (analysis) · Power BI (dashboard)
Core question	Which factors — rating, price, popularity, location — are associated with a restaurant closing?

Executive Summary

This project traces one question through a complete analytics pipeline: what separates a restaurant that survives from one that closes? Starting from a **5GB raw Yelp data** export, the workflow moves through Python-based extraction and cleaning, PostgreSQL analysis using window functions and multi-dimensional grouping, and finishes in an interactive Power BI dashboard.

Across **49,853 U.S. restaurants**, roughly **one in three (33.3%) has closed**. The analysis finds that review volume , not star rating , is the strongest single predictor of survival: restaurants in the bottom quartile of review counts close more than twice as often as those in the top quartile. Star rating has a non-linear, and initially counter-intuitive, relationship with closure: restaurants with “mediocre” 3–3.5-star ratings close more often than 1-star restaurants. Price tier compounds the risk — expensive restaurants with average-or-below ratings are the highest-risk segment in the entire dataset.

Headline numbers

33.3% overall closure rate · 3.52 average star rating · 90.2 average reviews per restaurant · 49,853 restaurants across 13 states

1. Project Overview

1.1 Objective

Identify which measurable factors : **star rating, price range, review volume, and location -** are associated with a restaurant being closed rather than open, and package the findings as an interactive dashboard suitable for further exploration.

1.2 Data Source

Data was sourced from the **Yelp Open Dataset** (business.yelp.com/data/resources/open-dataset), a public academic dataset of business listings across multiple U.S. metro areas, distributed as a set of newline-delimited JSON files inside a single ~5GB archive.

1.3 Scope

Stage	Row count	Note
Raw businesses extracted	150,346	All business types — not restaurant-specific
Filtered to restaurants	52,268	categories LIKE '%Restaurants%'
Final analysis dataset	49,853	After removing Alberta (non-U.S.) and noise states

2. Data Pipeline

The raw Yelp export was too large to load into memory as a single file, which shaped several early decisions in the pipeline.

2.1 Extraction

The source .tar archive (~5GB) was streamed rather than fully extracted: a Python script using the tarfile module located and extracted only business. Json, avoiding the memory cost of unpacking the entire archive.

2.2 JSON to CSV Conversion

business. Json is newline-delimited JSON (one business record per line). Each line was parsed individually and written to CSV via csv.DictWriter, using the keys of the first record to define the column headers.

2.3 Column Reduction & Price Extraction

Six columns were dropped as unnecessary for this analysis: address, postal_code, latitude, longitude, categories, hours, and attributes. Before dropping attributes, a regex-based parser extracted the RestaurantsPriceRange2 value nested inside it (three fallback patterns were tried, covering different quoting styles), defaulting to price range 2 ($$) when no value was present.

Price tier	Count	% of businesses
1 ($)	28,840	19.2%
2 ($$)	113,647	75.6%
3 ($$$)	6,667	4.4%
4 ($$$$)	1,192	0.8%

2.4 Database Load

A restaurants table was created in PostgreSQL and all 150,346 cleaned rows were loaded via psycopg2, using ON CONFLICT (business_id) DO NOTHING to make the load idempotent against re-runs.

3. Data Cleaning & Scope Decisions

- Restaurant filter - narrowed 150,346 businesses to 52,268 rows where categories contained “Restaurants,” created as a dedicated restaurants_only table.
- Currency scope - removed 2,410 rows in Alberta, Canada (state = 'AB'), since Canadian pricing conventions would distort price-range comparisons built for a U.S. market.
- Noise removal - removed states with fewer than 3 records (5 rows total across a handful of stray/erroneous state codes), which had no analytical value and would only clutter state-level rankings.

Final analysis dataset: 49,853 restaurants across 13 U.S. states and 903 cities.

4. Methodology: SQL Analysis

Nine SQL queries were built to progressively answer the core question, moving from a single baseline number to multi-dimensional and window-function-based comparisons. Each query's output is summarized in the Key Findings section that follows.

#	Query	Technique	Purpose
1	Baseline closure rate	COUNT / SUM / CASE	Establish the national average to compare every other result against
2	Closure by star rating	CASE WHEN bucketing	Test rating as a predictor of closure
3	Closure by price range	GROUP BY	Test price tier as a predictor of closure
4	Closure by review volume	NTILE (4) window function	Test popularity as a predictor, controlling for skew
5	Closure by state	GROUP BY	Compare geographic risk
6	Price × rating matrix	Two-dimensional GROUP BY	Combined risk surface (heatmap source)
7	Riskiest cities	GROUP BY + HAVING	Rank cities while filtering out small-sample noise
8	State vs. national average	AVG() OVER, RANK()	Relative risk ranking by state
9	Master view	CREATE VIEW	Single clean source feeding the Power BI dashboard

5. Key Findings

5.1 Review volume is the strongest single predictor

Restaurants were split into four equal-sized groups by review count. The closure rate falls steadily and substantially as review volume rises, restaurants with the fewest reviews close more than twice as often as those with the most.

	review_quartile integer	min_reviews integer	max_reviews integer	restaurants bigint	closure_rate_pct numeric
1	1	5	14	12464	44.4
2	2	14	35	12463	35.6
3	3	35	93	12463	32.3
4	4	93	7568	12463	20.8

5.2 “Mediocre” restaurants close more than poorly rated ones

Star rating's relationship to closure is not a straight line. 1–1.5-star restaurants close less often (20.2%) than 3–3.5-star restaurants (39.1%), the highest closure rate of any rating band. Being rated poorly does not predict closure nearly as strongly as being rated unremarkably: a 1-star restaurant tends to have a small, committed customer base, while a 3-star restaurant has no strong reason to be chosen over a competitor.

	star_bucket text	restaurants bigint	closure_rate_pct numeric
1	1-1.5 (very lo...	1677	20.2
2	2-2.5 (low)	7384	31.6
3	3-3.5 (mid)	18464	39.1
4	4-4.5 (high)	20822	30.4
5	5 (top)	1506	26.1

5.3 Higher price tiers carry more closure risk

Closure rate rises with price tier, from 30.5% at the cheapest tier to 47.1% at the third tier — a pattern consistent with higher fixed costs and thinner margins for error at higher price points.

	price_range integer	restaurants bigint	closure_rate_pct numeric
1	1	19146	30.5
2	2	29016	34.4
3	3	1513	47.1
4	4	178	40.4

5.4 Price and rating combine — expensive + mediocre is the danger zone

Looking at price and rating together sharpens the picture. Cheap restaurants can survive a poor rating (price tier 1, 1–1.5 stars close at only 15.4%) — customers accept the risk at low cost. Expensive restaurants cannot: at price tier 3, closure rates rise as high as 58.7–67.8% for 2–3.5-star restaurants (sample sizes of 118–600, large enough to trust). The single highest cell in the matrix, 87.5% for tier-3/1–1.5-star restaurants, is based on only 8 restaurants and should be read as noise, not signal.

Price tier	Star bucket	n	Closure rate
1 ($)	1–1.5	1,325	15.4%
1 ($)	5 (top)	382	42.9%
2 ($$)	2–2.5	3,208	43.7%
2 ($$)	5 (top)	1,111	20.2%
3 ($$$)	2–2.5	118	67.8%
3 ($$$)	3–3.5	600	58.7%
3 ($$$)	1–1.5	8	87.5% — small sample, treat as noise

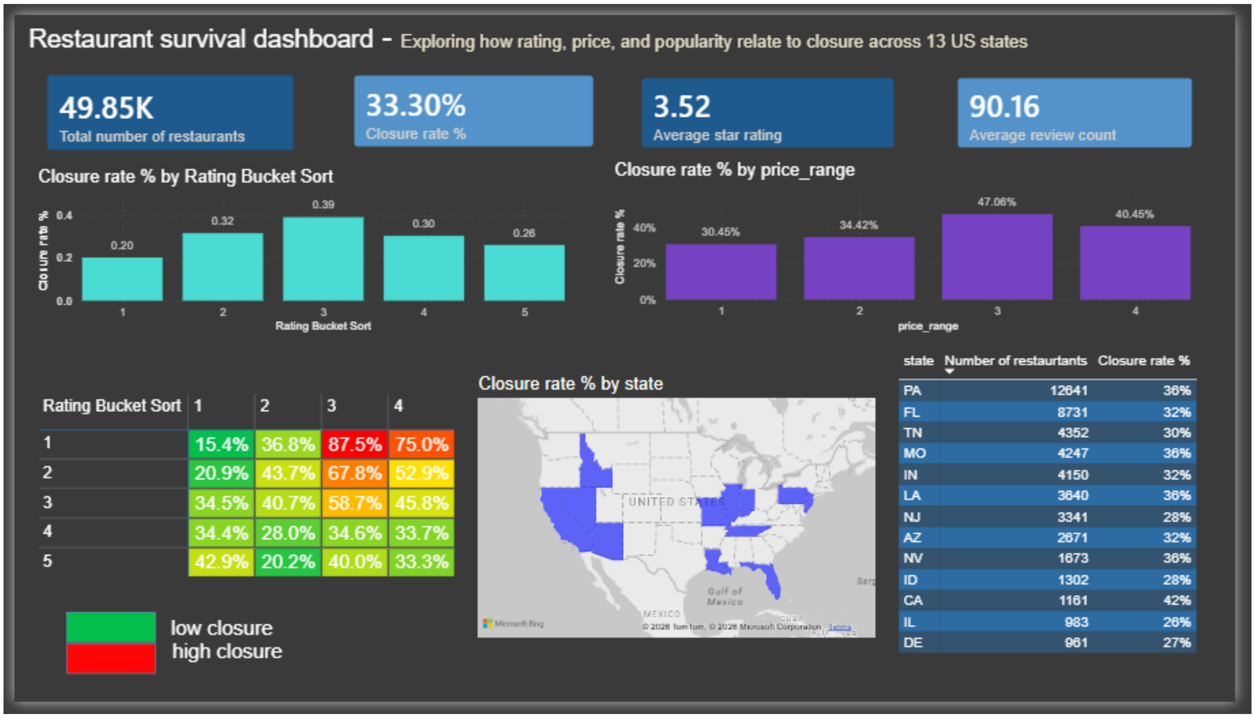
5.5 Riskiest individual cities (minimum 30 restaurants)

Filtering out cities with too few restaurants to be statistically meaningful (fewer than 30), Isla Vista, CA and Tampa Bay, FL top the list, with several Philadelphia-area suburbs (Chadds Ford, King of Prussia, Wayne, Bryn Mawr) also appearing — consistent with Pennsylvania's above-average state-level closure rate.

Showing rows: 1 to 10

	city character varying (100)	state character varying (10)	restaurants bigint	closure_rate_pct numeric
1	Isla Vista	CA	55	61.8
2	Tampa Bay	FL	36	55.6
3	Chadds Ford	PA	41	53.7
4	Clayton	MO	101	53.5
5	King of Prussia	PA	153	47.7
6	Wayne	PA	128	47.7
7	Goleta	CA	220	47.3
8	Gulfport	FL	55	47.3
9	Saint Louis	MO	1790	46.5
10	Bryn Mawr	PA	93	46.2

6. Power BI Dashboard



The nine SQL queries feed a single Power BI dashboard, built around one master view (restaurant_analysis) rather than nine separate imported tables, so every visual filters and slices consistently from one source of truth.

6.1 Layout

- KPI row — four cards: Total Restaurants, Closure Rate %, Average Star Rating, Average Review Count.
- Comparison row — closure rate by star bucket (bar chart), by price range (bar chart), and by state (filled map).
- Risk detail row — the price × rating matrix as a conditionally-formatted heatmap (green = low closure, red = high), and a table of the top 10 riskiest cities filtered to a 30-restaurant minimum sample.

6.2 Validation

The dashboard's live KPI cards — 33.30% closure rate, 3.52 average star rating, and 90.16 average review count — match this report's independently recomputed figures from the cleaned dataset, confirming the Power BI model is correctly connected to the final, cleaned data.

7. Technical Challenges & Learnings

- Large file handling — the 5GB source archive could not be loaded in full; targeted tarfile extraction of a single member file kept memory use low.
- Semi-structured data parsing — price range was nested inside a free-text attributes field with inconsistent quoting; a multi-pattern regex with a safe default (price range 2) handled the inconsistency without dropping rows.
- Currency/geography scope — recognizing that a Canadian province (Alberta) in an otherwise U.S. dataset would distort price and rent-style comparisons and removing it before analysis rather than after.
- Statistical noise control — using HAVING COUNT(*) >= 30 to prevent tiny-sample cities from dominating the "riskiest city" ranking with misleading 100% closure rates.
- Power BI measure vs. column bug — an early version of the dashboard summed a raw is_closed column instead of using a proper DIVIDE()-based measure, producing closure "rates" in the thousands of percent for larger cities. Rebuilding it as a measure that recalculates per filter context fixed the issue.
- Category sort order — text labels like "1–1.5 (very low)" do not reliably sort alphabetically; a numeric sort-by-column field was added so rating buckets always display in true low-to-high order.

8. Conclusion

Across **49,853** U.S. restaurants, closure is best predicted not by any single factor but by a combination: **low review volume is the strongest individual signal**, but the pattern sharpens further when price and rating are viewed together — **restaurants can survive being cheap and poorly rated but not being expensive and merely average**. Mid-tier ratings, not the lowest ratings, mark the riskiest segment, a genuinely counter-intuitive finding that a single-variable analysis would have missed.